

A Component Fidelity Audit

table 8 is the component-by-component audit summarised in Section 8.1, grounded in the code at tag `v2-paper-eval` and in the frozen Experiment 2 result JSON. It is reproduced in full because the three-way classification—defective, never exercised, contributing—is only checkable if the per-component evidence is visible.

Two notes on reading this table. First, the “unknown” 34.5% of fragments in Experiment 2—fragments whose originating leg could not be reconstructed from the stored records—is a logging limitation, not a fourth category of component; those fragments came from the legs that were running. Second, the entity-detection question is worth isolating because it is easy to get backwards: the pilot (Appendix B) genuinely used a regex tagger and had effectively no usable graph, but Experiment 2 loaded a trained model at roughly 95% recall. The mature system’s weak graph contribution is a chunk-size and representation problem, not a detection problem, and conflating the two would misdirect the repair.

B Experiment 1: The Pilot

System under test. The pilot evaluated an immature build with only the vector and BM25 legs meaningfully active. The classifier used a frozen MiniLM backbone; retrieval used static per-leg limits, a static 5,000-token global budget, and an uncapped 10-turn sliding window; the knowledge graph effectively did not exist—a regex tagger and an under-powered 1.5B extractor produced almost no usable edges—so the graph, procedural, and document legs contributed nothing and the system reduced to essentially the same vector-plus-BM25 retrieval as the baseline; cluster-scoped retrieval did not exist yet.

Protocol instantiation. Conversations shorter than 10 turns were excluded, reducing the corpus from 42 to 18. Split points were drawn from $[\max(10, 0.30L), 0.95L]$, three per conversation with a fixed seed; each split defined a historical block and a 10-turn future reference block. Probes were hand-written, 8–30 per conversation, and probes from earlier splits were re-asked at later ones. Ground truth used a single retrieval-assisted pass, frozen thereafter. Six conditions ran per probe.

The token-accounting audit. After the pilot completed, an audit found that the reported context size for full ICE included retrieved fragments but *omitted* the recent-turn sliding window incorporated during prompt assembly. Generated answers, scores, rankings, and hallucination labels were unaffected—the window was correctly supplied to the model at inference time; only the reported efficiency statistics were wrong. We replayed the sliding-window assembly for every checkpoint and added the omitted tokens. The correction did not change the qualitative findings, but it invalidated any claim of superior token efficiency: corrected, full ICE consumes *more* total tokens than the baseline at roughly equal quality. All pilot numbers reported here are corrected values.

The headline numbers were a defeat: 4.04 against 4.06, 47% more tokens, more hallucination, a worse TUR. The picture brightened only on the longitudinal axis—193 tracked question curves showed scores improving over time on repeated probes, demonstrating memory accumulation. Several subsystems were broken or immature: only three decay cycles had run; the graph effectively did not exist; 22 gating failures meant the classifier was actively blocking retrieval on probes that needed it; and the budget was static.

Lessons. A system that has not had time to consolidate memory is not a fair test of a memory architecture. Honest token counting is essential—the under-count would have made ICE look more efficient than it was. And a broken subsystem can drag down an entire

Table 8. Component state at the evaluation snapshot. “Contributing” means the component executed and affected retrieved output; “never exercised” means the benchmark supplied no input that would engage it; “defective” means it executed and failed.

Component	State	Evidence	Claimable as
Vector leg (decay-weighted)	Contributing	Typed vector bind present; dominates leg contributions	Validated
BM25 leg	Contributing	tsvector leg, no embedding bind needed; with vector, 62.3% of fragments	Validated
RRF fusion	Contributing	+0.82 [+0.39, +1.24] over a lexical and a dense leg	Validated, scoped to two legs
Dynamic token budget	Contributing	Stress test: 94.2% failure without, mean 4.33 with	Validated; strongest single result
Post-fusion curation	Contributing	32% fewer fragments at equal quality; SPF +50%	Validated in aggregate
Decay + reinforcement	Contributing	Simulated decay days per conversation (93/24/27/20); access-count and score restoration on every hit	Running; size not isolated
Cluster scoping	Contributing	Correct bind; ablation $\approx +0.01$	Measured neutral
Session diversify, dedup, bonuses	Contributing	Post-fusion transforms; keyword boost +0.12	Contributing
Codex graph	Contributing, under- weighted	3.3% of fragments, one concatenated fragment per probe; NER loaded and running at $\sim 95\%$ recall	Floor on content share; <i>not</i> an entity-detection failure
Classifier + gate	Contributing	Gating failures $22 \rightarrow 2$	Validated, with the override caveat
MoE routing	Measured neutral	Ran on every probe; $\Delta +0.01 /$ -0.02	Genuine null
Procedural leg	Defective	Untyped array bind $\Rightarrow$ vector $\Leftrightarrow$ double precision [] $\Rightarrow$ rollback $\Rightarrow$ []; 0.0 fragments	Bug; no claim
Document (RAG) leg	Never exercised	No documents ingested; store empty; leg also shared the bind defect	No claim either way
Batch summaries	Never exercised	Bind correct; nothing decayed far enough to batch; store empty	No claim either way
Cold storage	Never exercised	Replay horizon never reached the archival regime	No claim either way
Temporal / timeline retrieval	Never exercised	All 1,211 probes ask for current truth	No claim either way
Cross- conversation retrieval	Never exercised	Every probe scoped to one conversation	No claim either way
HyDE rewrite	Never isolated	Exp 2: disabled. Exp 3: gated by context-reliance, so active in <i>all</i> arms including bare vector; the flag toggled nothing	No claim either way

Table 9. How the two LSREP instantiations differ. The shared skeleton—state reconstruction, synchronous worker trigger, output-level judging by the same independent judge—is described in Section 5.2.

Dimension	Experiment 1 (pilot)	Experiment 2 (mature)
System maturity	Vector + BM25 legs; MiniLM classifier; static budget; uncapped sliding window; graph non-functional	All legs enabled; Qwen3 classifier + routing override; dynamic budget; cluster-scoped retrieval
Corpus	18 conversations (isolated checkpoints)	4 long-horizon conversations (full-lifespan replay)
Split range	[max(10, 0.30L), 0.95L], 3 splits/conv	Length-adaptive checkpoints (8–20), even spacing with jitter
Probes	Hand-written (8–30/conv)	Auto-generated, leg-aware (147) plus 72 inherited hand-written
Ground truth	Single retrieval-assisted pass, <i>frozen</i>	Three-stage forensic regeneration + forward propagation, <i>evolving</i>
Checkpoint state	Independent per checkpoint	Preserved across checkpoints; lifecycle runs between them
Probe evaluation	Each probe once	Incremental: re-scored at every later checkpoint
Scoring	Static correctness	Temporal-aware + inference credit; hallucination flags audited for false positives
Conditions	Six (incl. sliding-window controls)	Four (controls dropped)

Table 10. Experiment 1 (immature system), 657 probes, corrected token counts. After correcting the sliding-window under-count, ICE used more tokens than the vector baseline, scored slightly lower, hallucinated more, and had a worse TUR. The system was not yet a fair test of the architecture.

Condition	Score	Tokens	Hall. (%)	TUR	Win %
Control baseline (generalist)	3.05 ± 1.61	3,998	74.1	0.76	11.3
Control MoE	3.01 ± 1.61	3,998	74.4	0.75	7.6
Vector RAG (generalist)	4.06 ± 1.31	5,847	64.7	0.69	22.9
Vector RAG (MoE)	4.05 ± 1.32	5,847	66.7	0.69	15.6
Full ICE (generalist)	4.04 ± 1.33	8,586	69.6	0.47	22.0
Full ICE (MoE)	3.96 ± 1.33	8,586	72.3	0.46	20.6

system, which is itself a finding about the fragility of multi-component architectures and a direct ancestor of the fidelity audit in Appendix A.

C Full Ablation Results

Headline contrasts under the same paired bootstrap: the BM25 damage is -0.74 $[-1.12, -0.36]$ and the RRF rescue $+0.82$ $[+0.39, +1.23]$, both excluding zero; RRF-versus-bare is $+0.09$ $[-0.18, +0.36]$, full ICE versus bare is $+0.09$ $[-0.24, +0.42]$, and full ICE versus the single-leg baseline is -0.06 $[-0.40, +0.27]$, all spanning zero.

Table 11. Cumulative feature addition on Dataset B (67 probes), full buildup. Paired within-probe bootstrap, 10,000 resamples. Only +BM25 and +RRF clear a 95% CI; every other step spans zero. Steps whose feature was defective, never exercised, or never isolated at the snapshot (+procedural, +batch summaries, +HyDE) are *not* interpretable as neutral findings, and +Codex is under-weighted by construction (Section 8.1). The vector baseline is a reference point, not a buildup step.

Step	Score	Step Δ	Paired Δ	95% CI	SPF	Tokens	Frgs
bare_vector	3.27	—	—	—	0.23	14,873	14.0
+BM25	2.52	−0.75	−0.74	[−1.14, −0.36]	0.17	14,868	15.2
+RRF	3.36	+0.84	+0.82	[+0.39, +1.24]	0.29	14,870	11.7
+HyDE	3.39	+0.03	+0.03	[−0.13, +0.21]	0.29	14,870	11.6
+cluster_restrict	3.40	+0.01	+0.01	[−0.16, +0.21]	0.30	14,911	11.3
+session_diversify	3.40	+0.00	+0.00	[−0.22, +0.22]	0.30	14,911	11.2
+codex	3.43	+0.03	+0.03	[−0.21, +0.25]	0.30	14,912	11.3
+MERA	3.22	−0.21	−0.21	[−0.43, +0.01]	0.28	14,912	11.4
+procedural	3.39	+0.17	+0.16	[−0.07, +0.40]	0.30	14,912	11.4
+batch_summary	3.41	+0.02	−0.02	[−0.24, +0.23]	0.30	14,912	11.3
+dynamic_budget	3.30	−0.11	−0.08	[−0.36, +0.18]	0.16	24,489	21.2
+sliding_window	3.32	+0.02	−0.02	[−0.24, +0.23]	0.16	27,763	21.3
+keyword_boost	3.44	+0.12	+0.08	[−0.22, +0.37]	0.23	27,769	15.1
full_ice	3.38	−0.06	−0.03	[−0.29, +0.23]	0.22	27,768	15.0
vector_baseline (ref.)	3.42	—	—	—	0.11	16,373	30.0

Two steps warrant individual comment. **MERA**, the enumeration fallback for category queries (−0.21, [−0.43, +0.01], not significant), fires only when entity extraction returns nothing; since detection was working, it seldom triggered, so this delta rests on a small effective sample and is a downstream indicator of extraction gaps rather than an independent evaluation. **Codex** (+0.03) was live but emitted its entire traversal as one fragment, so a near-zero step delta is expected regardless of graph quality.

Table 12. Recency effect by origin split (fact age). Probes grouped by the turn index at which they were created. Recency Δ is mean(late) − mean(early). The fused configurations show positive recency—answers improve as the conversation grows—while the single-leg baseline is slightly negative.

Condition	Early (0–400)	Mid (400–800)	Late (800–1200)	Recency Δ
bare_vector	3.00	2.83	3.42	+0.42
+BM25	2.20	3.05	2.49	+0.29
+RRF	3.00	2.83	3.54	+0.54
+keyword_boost	2.90	3.28	3.60	+0.70
full_ice	3.10	3.25	3.47	+0.37
vector_baseline	3.50	3.15	3.47	−0.03

D Experiment 2 Detailed Breakdowns

Domain matters. On Dataset A (creative writing) ICE’s win rate is more than double the baseline’s. On Dataset B, the largest conversation at 1,119 turns, the baseline edges ahead

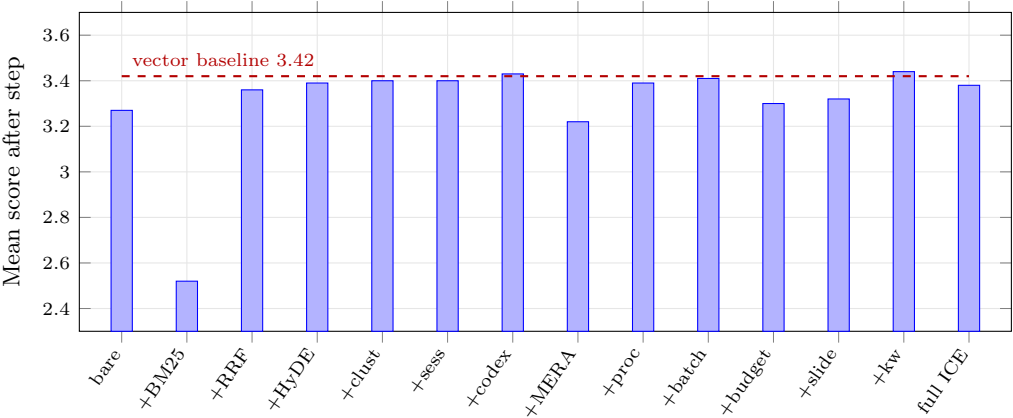


Fig. 3. Feature ablation buildup (Dataset B, 67 probes). Each bar is the cumulative mean score after adding one feature to all previous ones. The only two movements clearing a paired bootstrap CI are the +BM25 drop and the +RRF recovery. The dashed line marks the single-leg vector baseline, which is not a buildup step; the full build finishes level with it.

Table 13. Evaluation metrics. Each is computed per probe and aggregated across the benchmark.

Metric	Definition	What it measures
Score (1–5)	Mean absolute score from the judge	Answer quality
SPF	mean score/mean fragments	Retrieval efficiency
TUR	mean score/(mean tokens/1000)	Token efficiency
Win rate (%)	Blind tournament: all four conditions' answers for a probe are anonymised, shuffled, and ranked best-to-worst in a single pass, independent of absolute scoring; the win rate is the share of appearances placed first	Relative preference, free of judge order bias
Hallucination (%)	Share of probes where the judge detects information not present in the conversation	Faithfulness
Fragment count	Mean fragments injected per probe	Retrieval volume
Recency Δ	mean score(late bins) – mean score(early bins)	Longitudinal improvement

Table 14. Per-conversation breakdown (generalist conditions). ICE wins more head-to-head comparisons on every conversation. On Dataset A its win rate is more than double the baseline's.

Dataset	ICE score	Vec. score	ICE tok.	Vec. tok.	ICE win %	Vec. win %
A (Creative Writing)	3.63	3.50	19,434	24,360	37.2	18.0
B (Long-Form Creative)	4.28	4.33	24,108	21,257	32.8	22.9
D (Academic Planning)	4.63	4.57	20,103	18,076	20.4	18.8

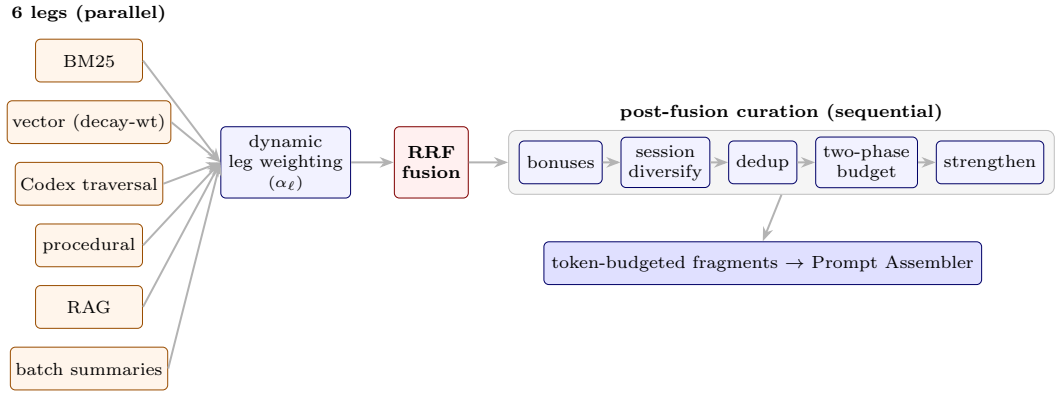


Fig. 4. Retrieval orchestrator. The legs are heterogeneous in retrieval semantics; RRF is what allows them to be combined without score normalisation; the two-phase budget guarantees leg diversity under budget pressure. Section 8.1 reports which legs contributed during evaluation.

on mean score (4.33 vs. 4.28) while ICE wins on win rate (32.8% vs. 22.9%) and uses 18% more tokens serving more detailed answers. On Dataset D ICE wins on both.

Table 15. Score distribution (1,057 probes, Dataset C excluded). The mean-score tie masks a structural difference: ICE-MoE has the lowest score-1 rate but shifts mass from 5 to 3–4, producing lower-variance output.

Score	ICE gen (%)	ICE MoE (%)	Vec. gen (%)	Vec. MoE (%)
1 (poor)	3.0	1.5	3.8	2.4
2	5.9	4.3	4.3	3.9
3 (acceptable)	13.3	16.6	14.8	19.5
4 (good)	18.1	19.7	17.2	16.3
5 (excellent)	59.7	58.0	60.0	58.0

Table 16. Temporal score quality (1,057 probes, Dataset C excluded). Buckets: Good (4–5), OK (3), Poor (1–2).

Condition	Good (4–5) %	OK (3) %	Poor (1–2) %
Vector RAG generalist	77.2	14.8	8.0
Vector RAG MoE	74.3	19.5	6.2
Full ICE generalist	77.8	13.3	8.9
Full ICE MoE	77.7	16.6	5.8

E Implementation Reference

This appendix collects operational detail supporting Section 4 but not required to follow the argument. A complete technical reference—the full training pipeline for the classifier and NER models, the controlled relation vocabulary in its entirety, GPU resource management,

Table 17. The four memory stores and the overlays. Each store is queried by one or more retrieval legs. Appendix E gives schemas and write paths.

Store	Purpose	Retrieval method
Episodic	Every conversational turn: raw text, summary, lossless flag, decay score, access count, embedding	BM25 leg; vector leg with decay weighting; session diversification
Codex	Temporally-versioned semantic graph: entities with aliases and properties, typed edges carrying <code>valid_from/valid_until</code> , append-only event log, snapshots	Graph traversal (BFS depth 3 over edges where <code>valid_until</code> IS NULL); enumeration fallback
Procedural	Recurring behavioural patterns: trigger conditions, reinforcement count, confidence	Vector top-5 behind a hard intent gate
RAG	Externally-ingested documents, chunked and embedded	Vector top-5, triple-gated
Memory slots	Seven server-enforced persistent slots (persona, preferences, project context, guidance, pending items, ...)	Prepended verbatim to every prompt
Context clusters	Conversation-scoped topical groupings with unit-norm centroids	Restrict episodic search to relevant clusters

and the idempotency architecture—is the archived technical report in the repository, which describes the system at the tagged evaluation snapshot.

E.1 Classification Engine

The engine produces, per user turn, a set of topic tags, a set of intent tags, and one context-reliance label. These drive every downstream decision: which legs are weighted, how the budget is split, whether the wide-net fallback fires, and which model the router selects.

The learned head is deliberately tiny—a $384 \rightarrow 128 \rightarrow 25$ multi-layer perceptron with ReLU and 0.3 dropout. Its input is a single 384-dimensional embedding from a frozen `Qwen3-Embedding-0.6B` encoder; the shared linear head is sliced at inference into three blocks: 11 topic labels, 11 intent labels, and 3 context-reliance classes. Topic and intent decode multi-label (sigmoid, threshold 0.3, argmax fallback); context-reliance decodes single-label.

The rule-based pre-classifier computes five density signals in $[0, 1]$ from the raw prompt—code density, sentiment density, meta density (references to the model itself), noise density, and reference density (anaphoric terms)—and evaluates five rules in order, returning the first that fires or none, in which case the learned head runs. The reference rule is special: it returns empty topic and intent tags but forces `Long_Term_Memory`, and the orchestrator then runs the head only to obtain topic and intent. A two-tier threshold on that rule (0.2 for short conversations, 0.1 past ten turns) is the engine’s primary long-conversation memory bias.

Override rules run in evaluation order after either path: memory immutability, so once `Long_Term_Memory` is set nothing may downgrade it; a topic rule promoting creative turns to `Long_Term_Memory`; and a rule promoting technical turns containing anaphora. A second, API-level bias lives outside the classifier: past ten turns, or below 0.95 confidence, a `Zero_Shot` label is upgraded before retrieval runs. Section 8 discusses what this costs.

Context-aware classification. When a conversation ID is supplied, the learned path queries the last three episodic turns for that conversation (max 500 words), preferring summaries and falling back to the first 150 words of raw text, and prepends that context to the prompt before embedding.

E.2 Codex Write Path and Retrieval

The graph spans four tables: entities (canonical name, aliases, properties JSONB, auto-regenerated context payload, embedding); edges (typed relations with strength, a confidence flag in {pending, active}, `valid_from`, and `valid_until`, where NULL means currently true); an append-only event log; and snapshots.

The triplet write path is the heart of temporal versioning. Entity resolution is two-stage—exact canonical-name match, then alias match, else create with a deterministic UUIDv5. The write branch then depends on the relation bucket: a property observation expires prior active edges of the same (source, relation) pair, writes a new edge, and overwrites the JSONB key; a reinforcement of an existing non-property edge increases strength and promotes from pending to active at strength ≥ 2 ; a new relation between the same pair expires the old one only if the old relation is not multi-valued. Every state change emits an event, and the entity’s context payload is rebuilt from the property map plus the ten most recent active outgoing edges. The event log is compacted by a worker that snapshots any entity exceeding 100 uncompact events—textbook event sourcing, with the live edge table as current state, the log as audit trail, and snapshots as bounded replay.

Micro-NER is a BIO tagger (a $384 \rightarrow 128 \rightarrow 64 \rightarrow 3$ MLP over per-token embeddings). When the trained model is unavailable the system falls back to a capitalisation regex minus a stoplist; at the Experiment 2 snapshot the trained model was present and loaded. **Vector fuzzy matching** embeds the extracted entity strings, scans entity embeddings, and takes the best above a cosine threshold of 0.85—the mechanism that resolves a slightly misspelled proper noun to its canonical entity.

The retrieval leg extracts prompt entities, resolves them by fuzzy matching, optionally restricts to a conversation scope, and performs a BFS traversal to depth 3 over edges where `valid_until` IS NULL, appending a labelled context payload per visited entity. **MERA**, the enumeration fallback for category queries, activates only when NER extracts no entities *and* the prompt contains both a category trigger and an enumeration hint; candidates are deduplicated and ranked by 30-day mention count.

E.3 Codex Limitations in Detail

The chunking paradox. Extraction used 6,000-token chunks with 200-token overlap, chosen to avoid splitting sentences and to preserve narrative context. A 4B-parameter model does not have the effective reasoning capacity to process 6,000 tokens at once; beyond roughly 1,000–1,500 tokens attention dilutes, losing entities mentioned mid-chunk while over-weighting the beginning and end. Two failure modes follow: *entity dropping*, where entities introduced mid-passage are never extracted, and *relationship hallucination*, where subject and object of distant clauses are confused, producing syntactically valid but meaningless triplets. The right chunk size for a 4B extractor is closer to 512–800 tokens.

Inert confidence and strength. Edges carry a confidence flag and a decaying numeric strength, but both are under-used at retrieval time. Traversal follows any edge with `valid_until` IS NULL regardless of either, and the only effect of an active confidence flag is a coarse $1.5\times$ fragment-level score boost. There is no reinforcement loop strengthening edges

on retrieval, so frequently referenced facts do not rise in the ranking and traversal treats weak edges as equal to strong ones.

Lexical versus semantic entity matching. Retrieval resolves entities by canonical name or alias plus fuzzy matching over the context-payload embedding rather than the entity name. A user who establishes “The Obsidian Citadel” and later asks “where is the main fortress located?” gives the resolution step no lexical or embedded anchor for “fortress,” though a human reader resolves it immediately. This is a general limitation of entity-centric retrieval in free-form conversation: it assumes users refer to entities by canonical names or close aliases, which natural dialogue does not honour.

E.4 Prompt Assembly

The assembler emits a message list in a deliberately stable-prefix order to maximise cache reuse: (i) a fixed system message with an inline persistent-context block rendering each active memory slot; (ii) recent turns as alternating user/assistant pairs under dynamic per-turn word caps; (iii) a single user message headed with the retrieved context, optionally tagged with cluster names when a cluster scope is populated; (iv) a single assistant acknowledgement acting as a boundary marker, so the model treats the final user message as the live question rather than another history turn; and (v) the live question. Because the system message, slots, and most of the recent-turn prefix change slowly, most of the prefix key/value tensors are in principle reusable across consecutive requests—though Section 8 reports that the practical benefit is small.

A two-layer budget enforces final size: the orchestrator packs fragments to the retrieval token ceiling, then the API layer recomputes total words against a per-model ceiling (2,770 words for the deployed model) and iteratively pops procedural and episodic fragments until the budget is met, preserving document and graph fragments because they are the hardest to recompute and the most likely to carry the answer.

Memory slots are prepended verbatim to every prompt. Seven slot names are server-enforced: persona, user preferences, tool guidelines, project context, guidance, pending items, and session patterns. Slots are written through four paths: direct user update, batch initialisation, reflection-proposed but user-gated updates (high-stakes slots land in a review queue), and reflection-applied updates to pending items only.

Context clusters are conversation-scoped topical groupings. The centroid of member turn embeddings, renormalised to unit length after every change, is a 384-dimensional vector; membership is many-to-many. The clustering worker assigns each unassigned turn to the best existing cluster—combining embedding similarity, a bonus per shared entity, and tag overlap—or opens a new cluster when no candidate clears a 0.6 similarity threshold.

E.5 Background Workers and Infrastructure

All GPU-touching workers gate on a utilisation threshold polled from the driver and, in shared-model mode, additionally on a user-activity key with a short idle window. Retry countdowns are fixed per class. All workers are idempotent through a unique idempotency key, turning at-least-once delivery into effectively-once side effects.

The system is a single service exposing an OpenAI-compatible chat-completions endpoint plus routers for memory slots, user control, and the model registry, backed by PostgreSQL with pgvector—one 384-dimensional vector column per vector table, because the embedder is shared—and a worker fleet over a message broker. Configuration is a typed settings object read from the environment. Server-sent event types (`classified`, `retrieval`, `context-ready`, `generating`, `degraded`) drive an observability panel and downstream monitoring.

Table 18. Background workers. The cluster is self-maintaining: no operator intervention is required for the memory lifecycle.

Worker	Trigger	Function
Post-flight evaluator	every turn	lossless detection, summary, dispatch extractors
Codex extractor	lossless turns	triplet extraction and versioned writes
Procedural extractor	every turn	pattern detection, reinforcement or insertion
Episodic decay	1.5 h	access-weighted decay, archive at 0.1, cold-store at 0.05
Codex decay	1.5 h	edge strength decay, demote below strength 0.3
Procedural decay	1.5 h	deactivation after 180 d with < 3 reinforcements
Clustering	30 min	assign unassigned turns, merge similar clusters
Reflection	2 h	session synthesis, pattern crystallisation, slot proposals
Batch summariser	2 h	50-turn batches, preservation-prompt summaries
Sentinel monitor	30 min	declarative rules: threshold, absence, and others
Fine-tune	weekly	retrain the classifier head on curated labels
Compaction	manual	snapshot entities with ≥ 100 uncompact events
Drop zone	filesystem	ingest documents into the document store
Codex inject watcher	filesystem	ingest structured files into the graph with manual confidence

Classifier fine-tuning writes a timestamped checkpoint but does not auto-promote it, so production promotion requires a deliberate swap.